

CANDIDATE
NAME

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PDG
INDEX NUMBER

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BIOLOGY

9648/03

Applications Paper and Planning Question
Paper 3

14 September 2015
Monday

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

All working for numerical answers must be shown.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Calculators may be used

For Examiner's Use	
1	
2	
3	
4	
5	
Total	72

Answer **all** the questions.

- 1 In 2008, a woman had her damaged left bronchus replaced by one constructed using her own stem cells. A section of the trachea from a donor was treated in order to remove the cells of the donor, leaving only a framework of collagen. Adult stem cells derived from the woman's own bone marrow were then added to this collagen framework. Samples of these stem cells, together with cells derived from the lining of her trachea, were placed in a bioreactor for four days during which time they multiplied to coat the collagen framework. This was used to replace the damaged section of her bronchus. A month later the transplanted tissue had developed its own blood supply. This was claimed to be the first successful transplant using tissues derived from the patient's own stem cells.

- (a) Outline the advantages of using a patient's own stem cells to treat a damaged bronchus rather than transplanting a complete bronchus.

- Patient's immune system will **not** be stimulated to **cause immune response resulting in tissue rejection**;
- (patient's own stem cell) does not contain **foreign antigens (glycoproteins)**;
- No need for immunosuppressant/ drug;
- Overcomes problem of finding suitable donors such as tissue/ blood matching / no need to decide who is eligible;
- Fewer ethical sourcing, issue than when using embryonic stem cells
- Bronchus cells can be cultured , in large quantities;
- Healthy culture of patient's stem cells can be stored for future needs;

[3]

- (b) Once most stem cells differentiate, they lost their ability to turn into other types of cells. However, some fully differentiated cells can be stimulated to change back into stem cells in tissue culture. Such cells are called induced pluripotent stem cells (iPS cells).

In an experiment with mice to produce iPS cells, it was discovered that the introduction of four genes would cause certain fully differentiated cells to change to iPS cells. These four genes indirectly result in the reprogramming of the differentiated cells by influencing other genes.

- (i) Suggest **two** possible means by which the four genes were introduced into the cells.

- Creation of (Cationic) liposomes which carries the DNA → Fuse with cell surface membrane/electroporation
- Direct introduction of DNA/ plasmids into nucleus via micropipette;
- Genetically modified adenoviruses enter via receptor-mediated endocytosis
- Genetically-modified retroviruses enter via membrane fusion;

[2]

- (ii) Outline how the addition of the four genes may influence the activity of many other genes to result in the reprogramming of cells.

- The **genes code for transcription factors / activators and repressors** respectively/**chromatin remodelling enzymes** that

[Give example]

- ...**increases the expression of genes** that control **pluripotency** of cells.
- AND
- ...**decreases the expression of genes** that resulted in **differentiation** of cells.

[2]

- (iii) One of the steps in the process of developing induced pluripotent stem cells is the need to determine whether the cell line is pluripotent. One assay that has been used to determine the state of pluripotency is a teratoma forming assay.

Teratoma are tumors of multiple lineages containing tissue derived from all three germ layers, which is unlike other tumors, which typically are of only one cell type.

Suggest why the formation of a teratoma is indicative of a pluripotent state.

- (define pluripotent) a cell must be able to differentiate into any cell type derived from the three germ layers (endoderm, mesoderm or ectoderm).
- If a teratoma forms, researchers can **confirm the presence** of these three tissue types within the teratoma. If the teratoma does produce tissues from all the germ layers, the cell line is considered to be pluripotent. If the teratoma does not have all three germ layers present, then the cell line is not considered to be pluripotent.

[2]

- (c) There is evidence to suggest that the introduction of the four genes mentioned in (b) caused an increase in the production of telomerase reverse transcriptase (TERT) in the fully differentiated cells.

Explain how TERT may help to change the cells back into stem cells.

- TERT increases length of telomere/maintain number of **telomere** repeat sequences at the end of chromosome;
- **Telomeres shorten** after each division;
- TERT **counteracts the limit** of the number of times chromosome replicate/ **escape Hayflick limit**;
- Cell does not enter replicative cell senescence/ apoptosis (N2009 ER)
- so that the cell line can continue to divide/ self-renew by mitosis infinitely/ repeatedly;

[3]

(d) A cybrid (cytoplasmic hybrid cell) is produced as shown in Fig. 1.1.

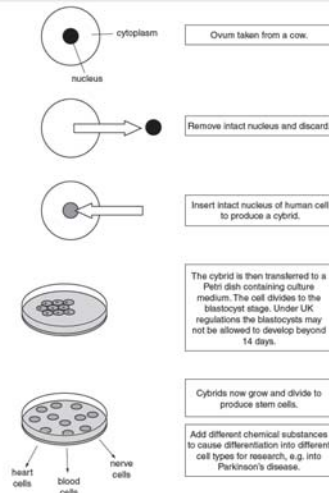


Fig. 1.1

- (i) The DNA of such a cybrid is 99.6% human. The remaining 0.4% of the DNA is found in the cytoplasm. Explain why there is DNA in the cytoplasm of the cybrid.

In **mitochondria** from **cow**

[1]

- (ii) When the Human Fertilisation and Embryology Bill was considered by the UK parliament in 2008, some people argued that it is unethical to allow the production of cybrids.

State whether you agree or disagree that this is unethical and explain why you reached this decision.

Unethical;

1. presence of human and animal DNA means it is partly human and partly cow/ a human-animal hybrid which would not happen in nature;
2. claims of the benefits of embryonic stem cell research are over0rated/ few (if any) examples of successful in medical field;
3. dangerous precedent/ thin end of the wedge argument/ current applications (may be benign) but will lead to abuse in future;
4. possibility of unforeseen (irreversible) consequences;
5. unnecessary because, there already are/ will be soon, alternative techniques (to the use of embryonic stem cells);
6. e.g. use of adult stem cells/ umbilical cord cells;

ethical;

1. the amount of non-human DNA is very small/ negligible;
2. protocol limits keeping 'embryo' to 14 days (to guard against abuse) so cannot develop into whole organism;
3. provides a more ethical alternative to the use of embryonic stem cells from humans;
4. can provide many more stem cells than is possible from embryonic stem cells;
5. it would be unethical not to use cybrids to relieve, human suffering/ e.g. disease such as Parkinson's disease;
6. techniques which offer alternatives to embryonic stem cells builds on work done initially on embryonic stem cells
7. idea that ovum has bovine proteins would be rejected at implantation;

Allow other reasonable arguments on either side. If candidate does not express an opinion, limit score to one mark unless attempt has been made to argue that it is a finely balanced matter. Ignore references to 'playing God' and 'against religion'

[2]

[Total: 15]

- 2 Golden Rice™ is a genetically modified form of rice that produces relatively large amounts of β carotene in the endosperm. β carotene is metabolised in the human body to produce Vitamin A.

Fig 2.1 shows the metabolic pathway by which β carotene is synthesised in plants, and the enzymes that catalyse each step of the pathway.

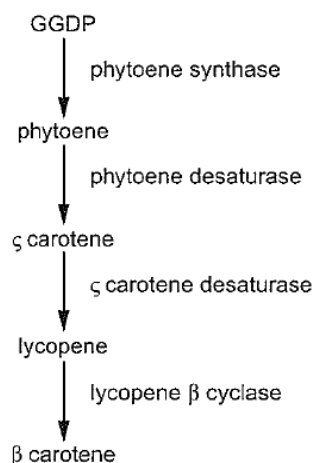


Fig. 2.1

The first types of Golden Rice™ contained a phytoene synthase gene, *psy*, from daffodils and a gene *crtI*, which produced the two desaturase enzymes, from the bacterium *Erwinia uredovora*.

It was found that the first types of Golden Rice™ produced only a very low mass of β carotene per gram of rice. Research continued to try to increase this.

Investigations were carried out to see if *psy* genes taken from species other than daffodils would enable rice endosperm to produce greater quantities of β carotene than the first types of Golden Rice™.

- *Psy* genes were isolated from the DNA of daffodil (Np), popcorn (Zm), chilli pepper (Ca), or tomato (Le). The genes were inserted into different plasmids.
- The promoter *Ubi1*, and *crtI* genes from *E. uredovora*, were also inserted into all of the plasmids.
- The four types of genetically modified plasmids were then inserted into different cultures of rice cells.
- The quantity of β carotene produced by these rice cells was measured.

Fig. 2.2 is a histogram showing the carotenoid content of maize calluses transformed with *psy* genes from different sources. C is the control plate without *psy* gene inserted into the maize calluses.

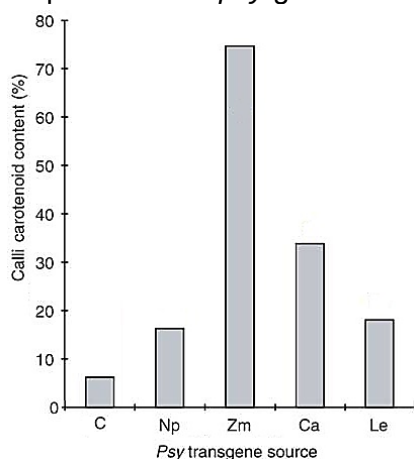


Fig 2.2

- (a) Using Fig. 2.1 and 2.2, suggest which step of the metabolic pathway was limiting the production of β carotene.
- Conversion of GGDP to phytoene step is limiting;
 - When *psy* gene from other plants were used, concentration of carotenoid in the cell increased;
 - Give comparison e.g. *psy* gene from popcorn increased cell carotenoid concentration by 57% compared to daffodil;
- [3]
- (b) With reference to Fig 2.2,
- (i) evaluate the effectiveness of the various *psy* genes in increasing the carotenoid content of the maize calluses;
- *psy* gene from popcorn is the most effective, followed by bell pepper, tomato and daffodil;
 - *data*: total carotenoid content is 75 % for popcorn, 35 % for bell pepper, only 15- 20 % for tomato and daffodil;
- [2]
- (ii) suggest an explanation for the differences in the carotenoid content of the maize calluses.
- **There may be slight differences in the *psy* gene sequence from the various source (must have);
 - resulting in a difference in the 3D conformation of phytoene synthase/ enzyme form giving rise to difference in affinity of substrate for the active site of the enzyme;
- OR
- So the stability of the enzyme form is different with some (from daffodil source) being degraded faster than others;
 - Add co-factor?
- [2]
- (c) Describe how a maize callus transformed with the plasmid containing the popcorn *psy* gene is formed.
- Explants are excised from the maize plant and surface sterilized in dilute bleach;
 - Explants are grown on culture media with a carbon source/ sucrose, organic nutrients/ amino acids, vitamins, inorganic ions/ minerals, and plant growth regulators;
 - During the process of tissue culture, plasmid containing the popcorn *psy* gene can be introduced to the plant cells using *Agrobacterium tumefaciens*/ gene gun;
 - In the presence of intermediate ratio of auxin and cytokinin the cells are allowed to divide mitotically, producing a mass of undifferentiated cells known as callus;
- [4]
- (d) Golden Rice™ are usually grown in isolated green houses away from conventional crops. Suggest reasons why this is so.
- Prevent genetic pollution – gene transferred to wild relatives, inherit traits from GM crops
 - Cross pollination may lead to drop in the quality of conventional and organic crop seed leading to a change in their performance and marketability or
 - Contamination of a conventional maize crop with GM maize may affect the market acceptability of the harvested crop due to reduced quality;

AVP

[2]

[Total: 13]

- 3 Restriction fragment length polymorphism, RFLP, refers to differences in fragment length generated by restriction enzymes. These differences are used to detect variations in genomes. The technique is widely used in molecular biology.

Fig 3.1 below shows an autosomal dominant disease segregating in a family. Genomic DNA from each family member was assayed for RFLPs for *Eco*RI sites, on Southern blots using probes derived from the gene.

The results for all individuals, except those for the foetus, are shown below the pedigree chart. The sizes of the DNA fragments obtained are as indicated.

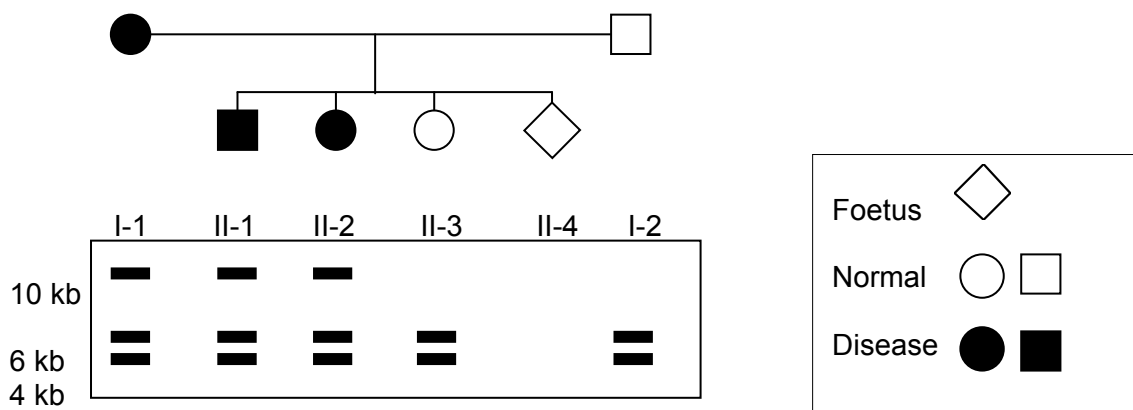


Fig 3.1

- (a) (i) Sketch a suitable diagram to illustrate the region of genomic DNA recognised by the probe in individual I-1. Show the positions of *Eco*RI sites and the distance between them in kb.

Answers



[Note: There should be 2 alleles since each individual is a diploid. Since the probe detects both alleles, diagram should include both allele 1 and 2.]

[1]

- (ii) Which RFLP fragment did the mother (I-1) contribute to the diseased children?
10 kb fragment

[1]

- (iii) What is the probability of the foetus having the disease? Explain your answer.
- 50% or $\frac{1}{2}$ probability
 - Mother's genotype is heterozygote; father's genotype is homozygous recessive so chance of receiving dominant allele $\frac{1}{2}$
- [Accept Punnett Square with defined alleles]

Genotype of foetus is either 10 kb (disease allele), 6 kb-4 kb OR 6 kb - 4kb, 6 kb – 4 kb

[2]

Four individuals, I to IV, were tested to see if they are related.

The possible restriction sites of a gene and the results of four individuals after subjecting their DNA to restriction enzyme digestion are shown in Fig. 3.2.

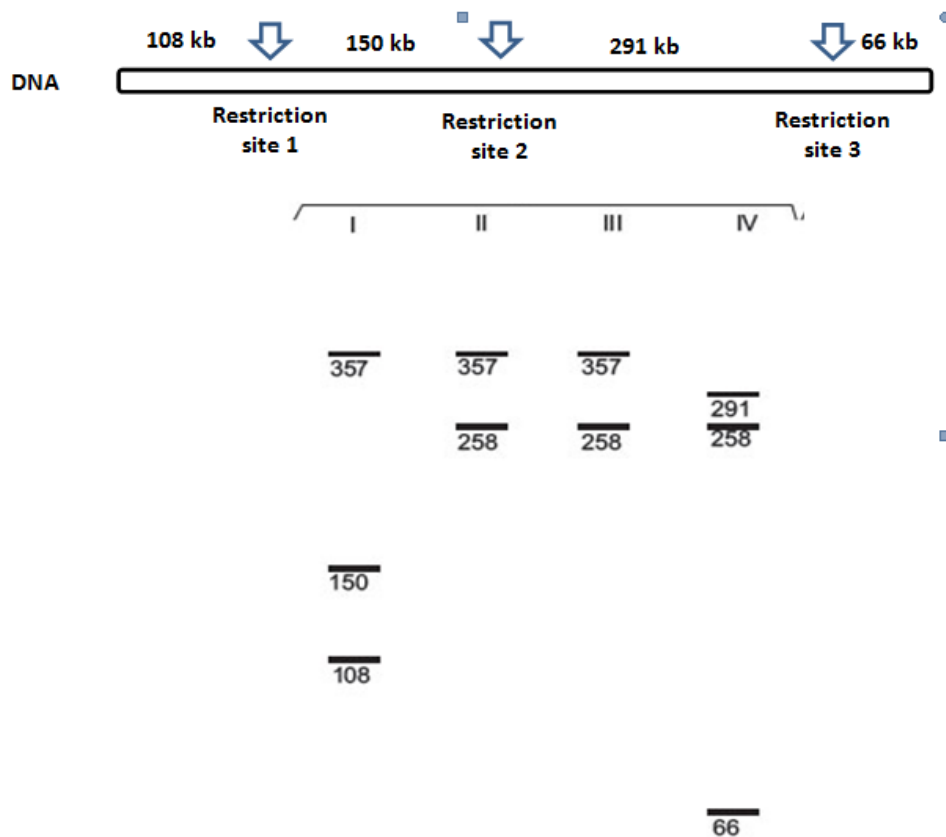


Fig. 3.2

(b) With reference to Fig. 3.2,

- (i) state how many alleles does this gene exists in these four individuals. Explain your answer.
- One allele has only restriction site 2 results in 357 kb and 259 kb fragments shown in individual II and III;
 - Another allele has restriction sites 1 and 2 resulting in 357 kb, 150 kb and 108 kb fragments in individual I;
 - Another allele has restriction sites 2 and 3 resulting in 357 kb, 150 kb and 108 kb fragments in individuals IV;

[3]

(iii) Suggest why this gene can exist in more than one allelic form.

- Single base substitution;
- Results in a loss of a restriction site,
- As restriction enzyme active site no longer complementary in shape to the restriction site shape once there is a change in a base respectively;

(Allows vice versa answer)

[3]

(iv) Explain why individuals II and III cannot be parents of individual I.

- Individual II and III have alleles containing only restriction site 2;
- Not possible for them to pass on to individual I, who have alleles containing restriction sites 1 and 2;

[2]

[Total:12]

4	Planning question P1 is a temperate phage that infects <i>Escherichia coli</i>. It is an unusual phage in that during the lysogenic stage, its DNA remain transcriptional silent and exists as an independent circular DNA plasmid-like element rather than integrates into the host genome like other temperate bacteria phage. In its lytic phase, P1 can produce transducing phages which carry part of the bacterial genome. Induction of P1 from the lysogenic cycle to the lytic cycle is known to occur over a temperature range of 32°C to 52°C. You are required to plan, but not carry out, an investigation to determine the optimum temperature at which induction of P1 phage from the lysogenic cycle into the lytic cycle occurs. Induction from the lysogenic to lytic cycle can be determined by observing the gene expression of β-galactosidase. β-galactosidase is an enzyme made up of two peptides, lac Z-α and lac Z-Ω, neither of which is active by itself but when both are present together, they spontaneously reassemble into a functional enzyme. In mutant <i>E. coli</i> with a deletion in the lac operon, β-galactosidase lacking the lac Z-α is produced. The mutation can be repaired by transduction with P1 transducing phages which have been previously been reproduced in <i>E. coli</i> that do not contain the deletion in lac operon. These transducing phages contain DNA encoding lac Z-α. Upon transduction of the mutant, lac operon deficient <i>E. coli</i> with transducing phages containing DNA encoding lac Z-α, the amount of β-galactosidase produced in the bacterial cell can be measured using a colorless chemical known as X-gal. Hydrolysis of X-gal by β-galactosidase will result in the formation of a blue product. The color intensity of the product can then be assayed using a colorimeter which measures the absorbance of different wavelengths of light in a solution in absorbance units (A.U). A solution of higher color intensity will absorb more light energy at a specific wavelength. Your plan should: • have an explanation of theory to support your practical procedure • have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it • include layout of results tables and graphs with clear headings and labels • include full details and explanations of the procedures that you would adopt to ensure that the results obtained were as quantitative, precise and reliable as possible • include relevant risks and precautions taken. [Total: 12] Your planning must be based on the assumption that you have been provided with the following equipment and materials, which you must use: • a glucose-deficient liquid culture of mutant <i>E. coli</i> previously infected with P1 transducing phages carrying DNA encoding for lac Z-α. The transduced mutant <i>E. coli</i> culture is kept at 30°C. P1 phages are known to exist as lysogens in mutant <i>E. coli</i> cells at 30°C. • lactose solution • thermostatically regulated waterbath • colorimeter with accompanying cuvettes • liquid X-gal • stopwatch • sterile test tubes • a variety of syringes for measuring volumes
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Theory and Hypothesis (2m)

- T1 During lysogenic cycle, presence of repressor protein prevents expression of genes/transcription of phage genes
- T2 transcription/ expression of lac- α (on phage DNA) occurs only during the lytic cycle;
- T3 At the optimum induction temperature, there are most number of phages which undergoes lytic cycle;
- T4 Highest amount of lac- α produced which assembles with lac- Ω to form most number highest amount of β -galactosidase
- T5 highest amount of X-gal hydrolysed (to produce more blue compound)
- T6 Bacterial culture has the highest color intensity $\rightarrow$ highest absorbance;

Method (6 m)

- ✓ Production of lac- Ω by bacteria cells during the lysogenic cycle

- M1 Add in a fixed/ numerical volume of lactose to a fixed /numerical volume of transduced mutant E.coli culture in a test tube using syringes;
- M2 Incubate at 30°C in a thermostatically regulated waterbath;
- M3 for a fixed period of time (20mins – 1 hour) time with a stopwatch;

Purpose:

- M4 alloactose will bind to lac repressor, causing the lac repressor to no longer bind operator;
- M5 RNA polymerase can bind to unblocked promoter $\rightarrow$ to allow the lac operon to transcribe and translate lac- Ω (which is required to assemble with lac- α to form functional β -galactosidase during the lytic cycle);

- ✓ Induction

- M6 Add in stated volume of X-gal to test tube containing lactose, transduced mutant E.coli culture
- M7 Incubate at stated temperature (32°C) in thermostatically regulated waterbath;
- M8 Incubate for a fixed period of time (accept 10 mins - 1hour) time using a stopwatch;

- ✓ Measuring amount of β -galactosidase produced

- M9 remove a small volume of mixture and place into a cuvette using syringe;
- M10 measure absorbance using colorimeter;
- M11 Perform 2 more replicates at each temperature to obtain average absorbance;
- M12 Repeat experiment at least 4 more temperatures (temperature range from 32°C to 52°C, even intervals);
- M13 Repeat entire experiment twice to ensure reproducibility;

- ✓ Use of apparatus and techniques

- M14 Mention use of thermostatically waterbath in M2 and M6;
- M15 Mention use of stopwatch in M3 and M7
- M16 Mention use of syringes and test tubes to measure -transfer volumes of liquid and hold liquid culture respectively in M1 and M8
- M17 Mention aseptic techniques to prevent contamination

Accept e.c.f for steps not mentioned in procedure

Award max of 4m only if leave out the procedure for production of lac- Ω by bacteria cells during the lysogenic cycle

Control (1m)

- C1 Control: transduced mutant E.coli culture + lactose incubated at 30°C subjected to the same experimental conditions;
- C2 Purpose: To show that transduced mutant E.coli produce functional β -galactosidase only upon induction of P1 phage into the lytic cycle;

Results (2m)

✓ Table

R1 correct headings with appropriate units;

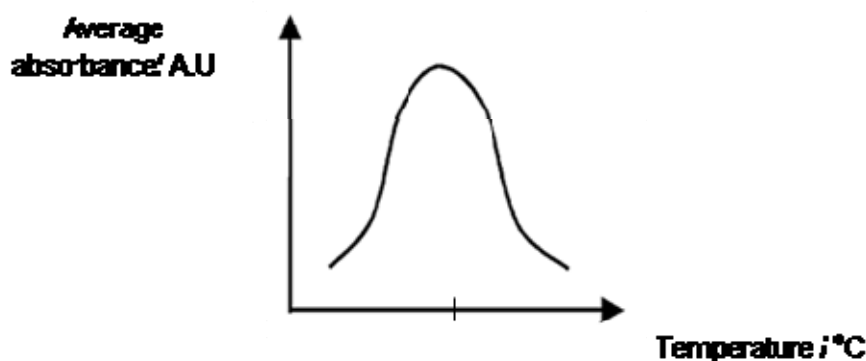
R2 Replicate and average included;

Temperature/ °C	Absorbance / A.U			
	Replicate 1	Replicate 2	Replicate 3	Average
32				
36				
40				
44				
48				
52				

✓ Graph

R3 x-axis: temperature / °C, y-axis: average absorbance / A.U
allow ecf from R1 and R2;

R4 Graph drawn with a peak at optimum temperature;



Safety (1m)

S1 Risk: Exposure/ contact with bacteria or bacteriophage may result in bacterial or viral infection

S2 Precaution: any 1

- Before starting work, cover all cut or broken skin with a waterproof dressing;
- Wear gloves;
- Before and after each working session clean the bench surface with 70% ethanol / swab any spillages with bactericidal disinfectant;
- All solutions and apparatus that have come into contact with bacteria and bacteriophage should be placed in fresh 10% bleach solution for sterilization before disposing them;

Free-response question

Write your answers to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5	(a)	Describe how RFLP can determine completely linked genes and how RFLP analysis can be used in genome mapping.	[6]
		Simpler version: Describe how RFLP analysis can be used in genome mapping. 1. Upon <u>restriction digest</u> on a particular region of DNA, RFLP results in varying <u>lengths of restriction fragments</u>; 2. Arises due to <u>different numbers of restriction enzyme recognition sites</u> or <u>variable number of tandem repeats</u> (VNTR/STR) between restriction sites; 3. RFLP analysis: involves amplification by <u>PCR (optional)</u> → cutting by <u>restriction enzymes</u> → separation by <u>gel electrophoresis</u> → 4. transfer by <u>Southern blotting + nucleic acid hybridization</u> with radioactive probe; 5. RFLP loci can be used as genetic markers to map other linked genes / other genetic marker 6. RFLP markers are <u>inherited in a Mendelian fashion</u>; 7. RFLP analysis can be performed on <u>RFLP markers located on two different loci</u> of the genome RFLP determine completely linked genes 8. <u>2 individuals</u>, each being <u>homozygous for each RFLP marker</u>, but have <u>different band patterns</u> for each particular RFLP marker, can be crossed to <u>produce heterozygous offsprings</u>; 9. <u>Heterozygotes</u> are then <u>crossed with one of the homozygous parent</u> 10. offspring can <u>inherit pairs of RFLP markers like the parents</u> or <u>inherit recombinant pairs of RFLP markers</u>; 11. If the 2 gene loci are <u>linked without crossing over</u>, there will only be <u>2 phenotypes</u> observed 12. as the <u>two gene loci will assort and segregate together</u>; RFLP analysis used in genome mapping. 13. If the 2 gene loci are <u>not linked</u>, number of offspring showing the four phenotypes will be of ratio <u>1:1:1:1</u>; 14. If the two gene loci are <u>closely linked</u>, the <u>proportion of offspring with recombinant phenotypes will be less than parental phenotypes</u>; 15. the frequency at which two RFLP markers or a RFLP marker and a certain allele of a gene undergo <u>crossing-over</u> is a measure of the <u>relative distance</u> between the two loci on a chromosome 16. RFLP loci which are <u>heterozygous</u> in individuals may be used as markers in chromosomal mapping by <u>crossing them</u>; 17. <u>1 % = 1 centiMorgan</u> OR the lower (higher) the frequency of crossing over/ recombination frequency between the two gene loci, the closer (further apart) the two loci 18. By performing <u>analysis on many other pairs of RFLP markers</u> and with other cytogenetic studies, the researcher is able to come up with a detailed map of the genome; 19. RFLPs can be mapped against gene loci once many RFLPs have been found/ RFLPs can be used as positions from which genes of interest can be found;	
	(b)	"Gene therapy clinical trials have mostly shown to be unsuccessful in terms of therapeutic interest." Explain why.	[7]
		1. Difficult to get DNA to <u>integrate</u> into the target cell genome 2. Results can be short-lived / patients need to be treated on a frequent basis 3. Problem with controlling the activity of <u>gene expression</u> 4. Risk of stimulating immune response	

	5. Virus vector may regain / develop virulence 6. Incorrect insertion of gene may cause <u>named / described problem</u> to patient 7. Possible problem with finding vector for large gene 8. Many genetic diseases are a result of presence of many defective genes / impossible to introduce many genes at same time 9. Problem dealing with, dominant condition / non dividing cells 10. Toxicity / low efficiency of liposomes 11. AVP	
(c)	Discuss the social and ethical concerns pertaining to the use of genetically modified animals.	[7]
	1 release (accidental or otherwise) of GM animals into the wild; 2 may result in GM animals out competing wild types; 3 ecological balance is disrupted / severe impacts on the food-chain; 4 destabilises & hence threatens biodiversity; 5 E.g. larger transgenic salmon may be preferably selected as mates over smaller wild types; 6 introduction of foreign gene(s); 7 may result in production of secondary metabolites; 8 maybe toxic to animals themselves and/or livestock/humans that consume them; 9 new proteins in GM animals; 10 may be potentially allergenic to humans that consume them; 11 animal rights – GM animals may suffer unnecessarily; 12 E.g. oncomouse; 13 which may increased susceptibility to cancer; 14 rearing animals for transplant purposes; 15 religious implications in food choice; 16 especially when GM foods are unlabelled; 17 E.g. incorporation of pig genes into cows; 18 ownership issues – unfair for large multinational companies to patent GM animals / increased dependence of undeveloped countries on rich developed countries	
	S	[Total: 20]